NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

26 October 2026

117 verified · 220 raw · 5 sources

117	2	115	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	110	Media
HackerNews	3	Community
TechCrunch AI	3	Media
The Verge AI	2	Media
MIT Tech Review	1	Media

VERIFIED INTELLIGENCE — 2 stories

VERI

Show HN: Junior – an open-source AI junior engineer that runs on your Mac

HackerNews +1 source 90% HN 4

Hi HN, I'm Samar, a software engineer. I built Junior because I was spending more time fixing CI issues, babysitting PRs and resolving merge conflicts. I thought maybe i should automate all of it, and that's why I started building Junior. Junior is an agent that runs locally on...

https://github.com/usejunior-sh/junior

VERI

Understanding Communication Failures in Multi-Agent LLM Systems

ArXiv CS.AI +1 source 80%

arXiv:2610.32060v1 Announce Type: new Abstract: Multi-agent LLM systems aim to improve reasoning through debate, evidence sharing, and mutual correction, but interaction can also introduce failures that final accuracy alone cannot explain. Agents may converge too early, ignore...

https://arxiv.org/abs/2610.32060

CONFIRMED SIGNALS — 115 stories

CONF

Show HN: CaseCite – Free RAG caselaw research and document analysis for lawyers

HackerNews 50% HN 14

It bugged me recently that the New York City subway countdown clocks show an estimated time that is often inaccurate. Since many of the delays are due to other trains on the same track, I figured: why not just show the trains? Source code:...

https://github.com/thomasunise/casecite

CONF

Show HN: God – real-time control room for LLM agent societies (EMNLP 2026 demo)

HackerNews 50% HN 2

No summary — see link for full article.

https://github.com/XiaoLuoLYG/GOD

CONF

AI delivers ‘zero financial payoff’ to most companies, Apollo economist says

TechCrunch AI 50%

"Despite massive investments and widespread hype around artificial intelligence, the technology has delivered zero financial payoff to corporate America outside of tech."

https://techcrunch.com/2026/10/25/ai-delivers-zero-financial-payoff-to-most-companies-a...

CONF

The hottest AI wearables and gadgets you should know about

TechCrunch AI 50%

We compiled a list of all the hottest AI wearables, including Muse Charm, Bee, Friend, Plaud, and more.

https://techcrunch.com/2026/10/25/the-hottest-ai-wearables-and-gadgets-you-should-know-...

CONF

Here are the top AI agents that can live in your text messages

TechCrunch AI 50%

We created a list of the most notable AI agents that can live in your text messages, from general assistants to agents designed for families, travel, and work.

https://techcrunch.com/2026/10/24/all-the-ai-agents-that-can-live-in-your-text-messages/

CONF

r/opensource just banned AI, non-English speakers, and more

The Verge AI 50%

The mods at r/opensource posted new submission rules that, among other things, banned projects that use AI. The post made Saturday evening opens by saying, "If your project uses AI, is made by AI, or is intended for AI, it's not welcome on r/opensource. AI, as it exists today,...

https://www.theverge.com/ai-artificial-intelligence/1018217/reddit-opensource-ai-ban

CONF

Google’s AI search changes upended online publishing

The Verge AI 50%

This is The Stepback, a weekly newsletter breaking down one essential story from the tech world. For more about AI's impact on the web, follow Emma Roth. The Stepback arrives in our subscribers' inboxes on Sunday at 8AM ET. Opt in for The Stepback here. How it started The...

https://www.theverge.com/column/1017983/google-ai-search-online-publishing

0 CONF

Don't Forget What You Just Learned: Self-Directed Repair Turns Agent Mistakes Into Lasting Lessons

ArXiv CS.AI 50%

arXiv:2610.31846v1 Announce Type: new Abstract: Intelligent agents should learn from their mistakes so that experience leads to continuous improvement. Yet today's agents recover from a mistake only temporarily: in-context reflections are easily forgotten across sessions and...

https://arxiv.org/abs/2610.31846

1 CONF

ToolRACER: A Robust Agentic Conversation Emulation Resource for Agent Training and Evaluation

ArXiv CS.AI 50%

arXiv:2610.09163v2 Announce Type: replace-cross Abstract: Task-oriented conversational agents remain fragile under real world conversation scenarios as they rarely follow a predictable script, especially when users exhibit non-cooperative behavior. Existing function-calling...

https://arxiv.org/abs/2610.09163

2 CONF

Beyond On-Policy vs. Off-Policy: At What Level Should LLM Agent Distillation Follow the Student?

ArXiv CS.AI 50%

arXiv:2610.31918v1 Announce Type: new Abstract: Distilling interactive LLM agents requires choosing between teacher-generated demonstrations and student-generated experience, corresponding broadly to off-policy and on-policy distillation. While both have proven effective, we...

https://arxiv.org/abs/2610.31918

3 CONF

M^3: A Multi-LLM System for Multi-Turn Jailbreaking as a Markov Decision Process

ArXiv CS.AI 50%

arXiv:2610.31937v1 Announce Type: new Abstract: Modern Large Language Models (LLMs) remain vulnerable to carefully orchestrated multi-turn interactions that progressively bypass safety mechanisms. However, many existing red-teaming approaches rely on heuristics or impose strong...

https://arxiv.org/abs/2610.31937

4 CONF

Knowing Whom to Trust is not Enough: The Trust--Agreement Gap in Repeated Multi-Party LLM Negotiation

ArXiv CS.AI 50%

arXiv:2610.31971v1 Announce Type: new Abstract: Large language model (LLM) agents increasingly participate in negotiations on behalf of individuals and organizations, yet they remain vulnerable to manipulation, particularly in repeated multi-party interactions. Classical...

https://arxiv.org/abs/2610.31971

5 CONF

The Helpful Agent Problem: When Emergent Cooperation Among LLM Agents Bypasses Intended Workflows

ArXiv CS.AI 50%

arXiv:2610.31985v1 Announce Type: new Abstract: LLM agents are increasingly deployed in collaborative settings where they pursue separate tasks while sharing resources and operating under review or oversight processes. Such settings create safety risks not only when agents are...

https://arxiv.org/abs/2610.31985

6 CONF

CEMIR vs. FISER: a comparative study of reasoning-based and enrichment-based approaches to pragmatic intent recognition

ArXiv CS.AI 50%

arXiv:2610.32022v1 Announce Type: new Abstract: Intent recognition in human-robot interaction requires agents to resolve ambiguous natural language instructions by inferring the speaker's underlying goals. Research has increasingly focused on two notable paradigms for addressing...

https://arxiv.org/abs/2610.32022

7 CONF

RSI-Opt: Recursive Self-Improvement Optimizer Agents For Model Training

ArXiv CS.AI 50%

arXiv:2610.32026v1 Announce Type: new Abstract: Conventional neural-network training typically follows a predefined optimization recipe, motivating a LLM agent that monitors training dynamics and adjusts optimizer settings as training progresses, but a fixed agent policy does...

https://arxiv.org/abs/2610.32026

8 CONF

POSEIDON: Scaling Training Environments for Scientific Agents from Code Repositories

ArXiv CS.AI 50%

arXiv:2610.32448v1 Announce Type: new Abstract: Training environments remain a major bottleneck for agents in data-driven scientific discovery. Existing collections of executable and verifiable training environments are small, and their construction remains difficult to scale....

https://arxiv.org/abs/2610.32448

9 CONF

Can Test-Time Memory Evolution Outperform Test-Time Training for Scientific Discovery?

ArXiv CS.AI 50%

arXiv:2610.32046v1 Announce Type: new Abstract: Recent \emph{test-time training} (TTT) frameworks have enabled open-weight large language models (LLMs) to outperform inference-only frameworks powered by frontier models on scientific discovery tasks. However, TTT introduces...

https://arxiv.org/abs/2610.32046

0 CONF

HyScale: Scaling Long-Context Training for Learned Sparse Attention

ArXiv CS.AI 50%

arXiv:2610.32057v1 Announce Type: new Abstract: We present HyScale, a long-context training system for learned sparse attention. Learned sparse attention is a promising architecture for million-token contexts, widely adopted in production since DeepSeek-V3.2. Selection, however,...

https://arxiv.org/abs/2610.32057

1 CONF

Behaviour analysis for large language model-driven agents

ArXiv CS.AI 50%

arXiv:2610.32087v1 Announce Type: new Abstract: Understanding language-agent behaviour requires learning how actions are organised across a trajectory and how this organisation relates to context and outcomes. We propose a general framework that connects data-driven behavioural...

https://arxiv.org/abs/2610.32087

2 CONF

DELA: Learning Controllable Edits for Diffusion Language Models

ArXiv CS.AI 50%

arXiv:2610.32096v1 Announce Type: new Abstract: Diffusion language models (DLMs) support bidirectional denoising and iterative refinement, but generation remains largely controlled by token-level masking schedules. This leaves draft-level refinement implicit and limits explicit...

https://arxiv.org/abs/2610.32096

3 CONF

MPQ: A Message-Passing View of Post-Training Quantization

ArXiv CS.AI 50%

arXiv:2610.32127v1 Announce Type: new Abstract: Post-training quantization (PTQ) is a key technique for efficiently deploying large language models, as it compresses weights into low-bit representations without retraining. Existing PTQ solvers, including GPTQ, LDLQ, and block...

https://arxiv.org/abs/2610.32127

4 CONF

Not All Information Is Worth Asking For: Decision-Informed Information Seeking with LLMs

ArXiv CS.AI 50%

arXiv:2610.32181v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly used to support decisions under incomplete information, where deciding what to look up and when to stop matters as much as the final answer. However, LLMs fall into an information trap:...

https://arxiv.org/abs/2610.32181

5 CONF

PEAR: Adaptive Advantage Reshaping for Perception and Exploration in Large Vision-Language Models

ArXiv CS.AI 50%

arXiv:2610.32216v1 Announce Type: new Abstract: Effective multimodal reasoning requires a model to reliably utilize visual information while sufficiently exploring diverse reasoning paths. To achieve this goal, prior work combines visual dependency and token entropy to reshape...

https://arxiv.org/abs/2610.32216

6 CONF

Deep Learning Warm Starts for Trajectory Optimization on the International Space Station

ArXiv CS.AI 50%

arXiv:2505.05588v4 Announce Type: replace-cross Abstract: Trajectory optimization is a cornerstone of modern robot autonomy, enabling systems to compute trajectories and controls in real-time while respecting safety and physical constraints. However, it has seen limited usage in...

https://arxiv.org/abs/2505.05588

7 CONF

Training an MoE LLM in MXFP4: How We Closed the Gap to BF16

ArXiv CS.AI 50%

arXiv:2610.32229v1 Announce Type: new Abstract: Training large language models (LLMs) in sub-8-bit arithmetic promises substantial gains in memory bandwidth and compute throughput, yet FP4 formats introduce quantization error severe enough to destabilize optimization if left...

https://arxiv.org/abs/2610.32229

8 CONF

Financial Context-Rich Strategy for Actor-Critic based Agents: A Deep Reinforcement Learning Approach for Stock Trading

ArXiv CS.AI 50%

arXiv:2610.32237v1 Announce Type: new Abstract: In modern financial markets, more than 3/4th of the large volume trade is executed by algorithmic trading based on advanced trading strategies to attain higher precision and speed than traditional human traders. The conventional...

https://arxiv.org/abs/2610.32237

9 CONF

MedCaseAgent: From Multimodal Clinical Evidence to Rare Disease Case Reports

ArXiv CS.AI 50%

arXiv:2610.32258v1 Announce Type: new Abstract: Although Large Language Models (LLMs) have demonstrated potential in clinical documentation, generating the rigorous, evidence-based reasoning required for unprecedented, atypical, and rare disease case reports remains a...

https://arxiv.org/abs/2610.32258

0 CONF

Trial by Trace: Benchmarking AI Agent Harnesses for Reliability and Failure Attribution

ArXiv CS.AI 50%

arXiv:2610.32285v1 Announce Type: new Abstract: LLM agents are increasingly used to solve real-world problems, and their effectiveness depends on both the underlying models and the harnesses that support their operation. However, most existing benchmarks focus on model...

https://arxiv.org/abs/2610.32285

1 CONF

DMAIC-Bench: From Self-Reflection to Self-Restraint for Agents in Multi-Stage Analytical Tasks

ArXiv CS.AI 50%

arXiv:2610.32305v1 Announce Type: new Abstract: Multi-stage analytical tasks require LLM agents to use intermediate evidence to decide whether to continue, redirect or stop. Self-reflection, in which an agent critiques its output, is widely studied, yet a critique does not show...

https://arxiv.org/abs/2610.32305

2 CONF

Beyond Gain: Measuring Skill Internalization for Agent Skill Retirement

ArXiv CS.AI 50%

arXiv:2610.32322v1 Announce Type: new Abstract: LLM agents retrieve skills from public hubs that hold up to about 100,000 entries and serve many models. As models improve, a hub must find the skills a given model has internalized, meaning it can do without them, and retire them...

https://arxiv.org/abs/2610.32322

3 CONF

SEVI: Safe Evidential Real-World Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2610.31829v1 Announce Type: cross Abstract: Real-world reinforcement learning has enabled contact-rich manipulation tasks to be trained directly on physical robots, but unsafe interactions during online learning remain a key barrier to practical deployment. We propose Safe...

https://arxiv.org/abs/2610.31829

4 CONF

EvoSkill v2: Fully Agentic Skill Evolution from Benchmark Adaptation to a Stateful Evolver

ArXiv CS.AI 50%

arXiv:2610.25931v2 Announce Type: replace Abstract: EvoSkill v2 is a three-agent system that edits the instructions (skills) an agent is given, leaving its model and harness intact. The user starts a run by describing a benchmark and supplying a grader, which a benchmark adaptor...

https://arxiv.org/abs/2610.25931

5 CONF

Adaptive Control of Scientific Instruments with Nonstationary Responses through Multimodal Agents

ArXiv CS.AI 50%

arXiv:2610.32370v1 Announce Type: new Abstract: Controlling scientific instruments requires parameters to be adjusted progressively in response to feedback from successive operations. When unobserved operating conditions evolve with operational history, responses become...

https://arxiv.org/abs/2610.32370

6 CONF

LeafSkill: Efficient Leaf-Level Action Grounding for Recursive LLM Agents

ArXiv CS.AI 50%

arXiv:2610.32412v1 Announce Type: new Abstract: Recursive large language model (LLM) agents solve long-horizon tasks through decomposition, execution, and environmental feedback. Yet after producing an atomic subtask, they often invoke the same large model again to translate a...

https://arxiv.org/abs/2610.32412

7 CONF

Async-RBench: Benchmarking Asynchronous Task Execution in LLM Agents

ArXiv CS.AI 50%

arXiv:2610.32429v1 Announce Type: new Abstract: Large language model (LLM) agents increasingly execute tasks asynchronously: a main agent continues working while its subagents run, and the task state may change before their results arrive. The main agent must decide which work...

https://arxiv.org/abs/2610.32429

8 CONF

ALLMTS: Automated LLM-based Medical Triage System

ArXiv CS.AI 50%

arXiv:2610.32469v1 Announce Type: new Abstract: Telephone triage is a safety critical task: from a brief intake note, a nurse must decide whether a caller needs an ambulance, an emergency department, an appointment, or self care at home. Queried directly, large language models...

https://arxiv.org/abs/2610.32469

9 CONF

CASE: Confusion-Aware Skill Evolution for Frozen Multimodal Emotion Models

ArXiv CS.AI 50%

arXiv:2610.32481v1 Announce Type: new Abstract: Multimodal large language models (MLLMs) have substantially advanced Multimodal Emotion Recognition (MER) by enabling fine-grained emotion understanding through reasoning over multimodal evidence. However, they still exhibit...

https://arxiv.org/abs/2610.32481

10 CONF

When Agents Fail: Model or Harness--Who's to Blame?

ArXiv CS.AI 50%

arXiv:2610.32483v1 Announce Type: new Abstract: LLM agents combine models with execution harnesses, so agentic benchmark performance reflects their joint behavior. We argue that harnesses should therefore be evaluated through their interactions with underlying models. We develop...

https://arxiv.org/abs/2610.32483

11 CONF

IGG: A Benchmark for Interactive GUI Grounding under Visibility Constraints

ArXiv CS.AI 50%

arXiv:2610.32536v1 Announce Type: new Abstract: GUI grounding benchmarks ask whether an agent can localize a target that is already visible in a static screenshot. Real interfaces often violate this assumption: a target may be off-screen, appear only after a state change or...

https://arxiv.org/abs/2610.32536

12 CONF

Beyond Channel Data: Configuration-Enriched Radio Foundation Models

ArXiv CS.AI 50%

arXiv:2610.32542v1 Announce Type: new Abstract: This paper proposes incorporating channel configuration (CFG) information into Radio Foundation Model (RFM) pre-training and demonstrates that the resulting CFG-enriched embeddings serve as effective inductive priors for downstream...

https://arxiv.org/abs/2610.32542

13 CONF

Do Language Models Trust Their Own Justifications? A Study on Functional Consistency

ArXiv CS.AI 50%

arXiv:2610.32665v1 Announce Type: new Abstract: Large Language Models (LLMs) have been widely adopted in text classification tasks, where they not only output class predictions but also generate explanations that highlight the tokens deemed most relevant for reaching the...

https://arxiv.org/abs/2610.32665

4 CONF

Agentic Hazard Mapper: Emulating Expert Map-Making with AI Agents for Risk Communication

ArXiv CS.AI 50%

arXiv:2610.32738v1 Announce Type: new Abstract: Hazard maps are essential tools for public risk communication, preparedness, and evacuation planning, yet effective cartographic design is not always ensured due to limited access to cartographic expertise across local governments...

https://arxiv.org/abs/2610.32738

5 CONF

ActionRAG: Evidence-Refining Graph Diffusion for Long-Term Memory in LLM Agents

ArXiv CS.AI 50%

arXiv:2610.32752v1 Announce Type: new Abstract: LLM agents need long-term memory that can recover precise and mutually consistent evidence from growing interaction histories, particularly when answers depend on relations among facts rather than isolated semantic matches. Current...

https://arxiv.org/abs/2610.32752

6 CONF

Reimagining Enterprise RAG for Multi-Agent Systems: A Knowledge Layer That Organizes Before It Retrieves

ArXiv CS.AI 50%

arXiv:2610.32851v1 Announce Type: new Abstract: Enterprises are deploying agentic AI across the software development lifecycle, and they need the behavioral logic of the systems they act on, not stakeholder documentation. A seven-agent reverse-engineering pipeline recovered this...

https://arxiv.org/abs/2610.32851

7 CONF

A Foundation Model for Policy In-Context Longitudinal Causal Effect Estimation

ArXiv CS.AI 50%

arXiv:2610.32863v1 Announce Type: new Abstract: Longitudinal causal effect estimation is complicated by time-varying confounding and treatment-confounder feedback loops. Existing methods generally require data-specific nuisance-model fitting, and many are tailored to...

https://arxiv.org/abs/2610.32863

8 CONF

Do "True" and "False" Add Up to One? Complement Consistency in LLM-Agent Opinion Simulations

ArXiv CS.AI 50%

arXiv:2610.32874v1 Announce Type: new Abstract: In LLM-agent opinion simulations, each agent reports the probability that a claim is true after seeing its own previous probability and the probabilities reported by its neighbors, and that number becomes its state in the next...

https://arxiv.org/abs/2610.32874

9 CONF

Advancing SVD-based LLM Compression via Layer-Wise Error Model Search

ArXiv CS.AI 50%

arXiv:2610.32878v1 Announce Type: new Abstract: Low-rank SVD-based compression offers a powerful strategy to reduce the computational costs of LLMs. However, existing methods face two key limitations: (i) global rank allocation, where uncalibrated error proxies fail to capture...

https://arxiv.org/abs/2610.32878

0 CONF

NgramRL: Accelerating RL Post-Training for Structured Generation via Rollout-Aware N-gram Speculative Decoding

ArXiv CS.AI 50%

arXiv:2610.32881v1 Announce Type: new Abstract: Reinforcement learning (RL) post-training improves the reasoning capabilities of large language models, but its cost is dominated by autoregressive rollout generation, which can exceed 70% of training time. Existing...

https://arxiv.org/abs/2610.32881

1 CONF

Closing the Modality Gap via Simplex-Constrained Representations

ArXiv CS.AI 50%

arXiv:2610.32890v1 Announce Type: new Abstract: While multimodal contrastive models like CLIP and CLAP successfully learn shared representations, their embeddings typically exhibit a persistent \emph{modality gap}. We propose addressing this separation through a purely geometric...

https://arxiv.org/abs/2610.32890

2 CONF

SemantEq: Faithful Bidirectional Equivalence of Formal Statements

ArXiv CS.AI 50%

arXiv:2610.32919v1 Announce Type: new Abstract: Statement autoformalization translates mathematics from natural language into a proof assistant such as Lean 4, and models now do it at scale. As provers get stronger, the burden of correctness shifts from the proof to the...

https://arxiv.org/abs/2610.32919

3 CONF

An Agentic Approach with Evidence-Grounded Verification for Multimodal Emotion Reasoning in Conversation

ArXiv CS.AI 50%

arXiv:2610.31685v1 Announce Type: cross Abstract: LLM-based multimodal emotion reasoning systems perform well on standard benchmarks, but improving them through instruction tuning can be computationally expensive. Agentic pipelines built from frozen models avoid this cost, but...

https://arxiv.org/abs/2610.31685

4 CONF

Before the Ticket: Grounding HPC User Support in Documentation and Resolved Tickets with an Agentic LLM Assistant

ArXiv CS.AI 50%

arXiv:2610.31703v1 Announce Type: cross Abstract: Most questions that reach an HPC help desk have already been answered in the center's user guides or in a ticket someone else filed last semester; users still file new tickets because the guides are hard to navigate and the...

https://arxiv.org/abs/2610.31703

5 CONF

MBABench-2.0: Evaluating AI Agents on Constructing Financial Spreadsheets Beyond Simple Verifiability

ArXiv CS.AI 50%

arXiv:2610.31711v1 Announce Type: cross Abstract: Spreadsheets are the lingua franca of modern finance, driving valuation, forecast, audit, and many other core workflows of the industry. As LLM agents advance to automate knowledge work, spreadsheet construction is a natural and...

https://arxiv.org/abs/2610.31711

6 CONF

Reuse, Rewrite, or Reject: Evidence-Constrained Trace Reuse for Repeated Mobile GUI Control

ArXiv CS.AI 50%

arXiv:2610.31712v1 Announce Type: cross Abstract: Mobile GUI agents frequently perform repetitive tasks with similar interaction procedures. Reusing previous executions can reduce redundant LLM inference and execution overhead, but variations in user instructions and execution...

https://arxiv.org/abs/2610.31712

7 CONF

What Open-Source Projects Tell Their Coding Agents: A Census of AGENTS.md and CLAUDE.md

ArXiv CS.AI 50%

arXiv:2610.31715v1 Announce Type: cross Abstract: Coding agents can be guided with instruction files in a repository. We examine this emerging practice through a census of popular, active GitHub repositories in thirteen languages, treating each repository's root AGENTS.md and...

https://arxiv.org/abs/2610.31715

8 CONF

ReMedI: Inference-Time Refinement and Self-Distillation for Controllable Molecular Optimization

ArXiv CS.AI 50%

arXiv:2610.32103v1 Announce Type: cross Abstract: In drug development, lead optimization must improve competing properties, such as increasing permeability and reducing toxicity, while limiting scaffold changes that could compromise the lead's potency. RL-based post-training,...

https://arxiv.org/abs/2610.32103

9 CONF

Autoregressive Multimodal Anomaly Detection with Large Language Models for Microservice Systems

ArXiv CS.AI 50%

arXiv:2610.31722v1 Announce Type: cross Abstract: Microservice systems have become a dominant architecture for large-scale cloud applications, enabling rapid deployment and elastic scaling. However, ensuring the reliability and availability of microservice systems critically...

https://arxiv.org/abs/2610.31722

0 CONF

Learning to Be Monitored: Causal Bounds and Intervention-Aware Post-Training Recipes for Chain-of-Thought Monitorability

ArXiv CS.AI 50%

arXiv:2610.31733v1 Announce Type: cross Abstract: Chain-of-thought (CoT) monitoring aims to detect potentially harmful behaviors in large language models (LLMs) from their CoTs. Intervention-based evaluations measure this property by asking whether a monitor can detect the role...

https://arxiv.org/abs/2610.31733

1 CONF

Generalized green and red concepts for artificial intelligence, big data, and information and communication technologies: Multidimensional framework and countermeasures

ArXiv CS.AI 50%

arXiv:2610.31739v1 Announce Type: cross Abstract: The rapid expansions of artificial intelligence (AI), big data (BD), and information and communication technologies (ICT) are driving substantial growth in electricity, water, and material demands, with data centers projected to...

https://arxiv.org/abs/2610.31739

2 CONF

Benchmarks Are Surprisingly Compaction-Friendly

ArXiv CS.AI 50%

arXiv:2610.31778v1 Announce Type: cross Abstract: Long horizon agentic tasks are now pushing LLM agents past their context window. To continue the task, the agent scaffold has to free up the context window, a process known as context compaction. Compaction is generally lossy by...

https://arxiv.org/abs/2610.31778

3 CONF

PrefO2O: Zero-Shot Preference-Based Reinforcement Learning via Offline-to-Online Latent Preference Transfer

ArXiv CS.AI 50%

arXiv:2610.31812v1 Announce Type: cross Abstract: Preference-based reinforcement learning (PbRL) has emerged as a promising paradigm for learning robot behaviors aligned with human intent. However, its scalability is fundamentally limited by the need to collect expensive human...

https://arxiv.org/abs/2610.31812

4 CONF

Machine-Readable Operator Manuals over the ISO 11783 Network for Offline, Multilingual LLM Assistance

ArXiv CS.AI 50%

arXiv:2610.31863v1 Announce Type: cross Abstract: If agricultural machinery is to support intelligent, offline assistance, then the system must access technical documentation dynamically. In this paper, the Semantic Synchronization of operator manuals over ISO 11783-based CAN...

https://arxiv.org/abs/2610.31863

5 CONF

Agentic AI Defense: Decode-Time Provenance Binding for MCP

ArXiv CS.AI 50%

arXiv:2610.31974v1 Announce Type: cross Abstract: Agentic AI systems using the Model Context Protocol (MCP) are vulnerable to indirect prompt injection through untrusted tool outputs. We demonstrate cross-modal trust laundering, in which instructions concealed in an...

https://arxiv.org/abs/2610.31974

6 CONF

Concept Acupuncturist: An Efficient and Precise Concept Erasure Method for Diffusion Models via Sparse Autoencoders

ArXiv CS.AI 50%

arXiv:2610.32059v1 Announce Type: cross Abstract: Diffusion-based image generation models produce high-quality images but may generate harmful or policy-violating content, motivating machine unlearning (MU) to selectively remove target concepts while preserving general...

https://arxiv.org/abs/2610.32059

7 CONF

Controlled Evaluation of EEG Foundation Models: Random Initialisation, Band-Power and Validation-Selected Probes

ArXiv CS.AI 50%

arXiv:2610.32067v1 Announce Type: cross Abstract: EEG foundation models report downstream gains, yet their protocols seldom include three controls. These are the same encoder untrained, spectral features fitted on the same data, and a layer picked without looking at test scores....

https://arxiv.org/abs/2610.32067

8 CONF

REACT: Representational Early Analysis and Control of Chain-of-Thought Trajectories

ArXiv CS.AI 50%

arXiv:2610.32113v1 Announce Type: cross Abstract: Chain-of-thought (CoT) generation offers an opportunity to intervene before a reasoning model completes its answer. But does the point at which risk is easiest to detect also offer the most effective control? We study this...

https://arxiv.org/abs/2610.32113

9 CONF

Tampered ML: Evaluating LLM Judgment of Machine Learning Research

ArXiv CS.AI 50%

arXiv:2610.32114v1 Announce Type: cross Abstract: One approach to accelerating progress on AI alignment is to automate parts of alignment research itself. Doing so requires systems that can perform reliably on hard-to-supervise research tasks, including identifying flaws in...

https://arxiv.org/abs/2610.32114

0 CONF

Beyond Data Schedule: Fast LLM Pretraining on Data Mixtures by Loss-Reweighting

ArXiv CS.AI 50%

arXiv:2610.32141v1 Announce Type: cross Abstract: Large language models often acquire specialized capabilities through successive pretraining stages, but these gains can come at the cost of forgetting earlier knowledge. In this work, we investigate whether such trade-offs can be...

https://arxiv.org/abs/2610.32141

1 CONF

The LLENS Platform for On-Premises Large Language Model Agents in a Japanese University Hospital: Development and Evaluation Study

ArXiv CS.AI 50%

arXiv:2610.32151v1 Announce Type: cross Abstract: Large language model agents that use tools could take on part of the electronic health record and desk work that occupies physicians' time, but in many hospitals the medical information systems run inside closed networks, so the...

https://arxiv.org/abs/2610.32151

2 CONF

Sparsity-driven emergence of functional modularity in multitask neural networks

ArXiv CS.AI 50%

arXiv:2610.32153v1 Announce Type: cross Abstract: Neural networks can learn multiple tasks, but how learning organizes shared computation into specialized and reusable components remains poorly understood. In this work, we show that sparsity inducing regularization promotes...

https://arxiv.org/abs/2610.32153

3 CONF

MG-REPA: Multi-Granularity Representation Alignment for Generative Target Speaker Extraction

ArXiv CS.AI 50%

arXiv:2610.32161v1 Announce Type: cross Abstract: Generative target speaker extraction (TSE) models achieve high-quality extraction, but suffer from considerably slow training convergence. Representation Alignment (REPA) accelerates diffusion transformer training by aligning...

https://arxiv.org/abs/2610.32161

4 CONF

Do VLAs Know They Will Fail? Interpretable Success and Failure Patterns in Physical AI Systems

ArXiv CS.AI 50%

arXiv:2610.32208v1 Announce Type: cross Abstract: Vision-Language-Action (VLA) models have emerged as powerful general-purpose robot policies, but their successes and failures remain difficult to diagnose and understand. In this work, we show that sparse autoencoder (SAE)...

https://arxiv.org/abs/2610.32208

5 CONF

Head Dimension Matters in Language Models for Long-Context Retrieval

ArXiv CS.AI 50%

arXiv:2610.32213v1 Announce Type: cross Abstract: Multi-head attention retrieves and combines context representations through query--key interactions and value aggregation, with head dimension constraining the per-head representation space. In long-context processing, it remains...

https://arxiv.org/abs/2610.32213

6 CONF

"Not" is Not Lost in Translation: A Causal Analysis of Negation Representations in Multilingual LLMs

ArXiv CS.AI 50%

arXiv:2610.32281v1 Announce Type: cross Abstract: Negation occurs in every language, yet its surface realization differs drastically. Despite this, multilingual large language models (LLMs) translate negation robustly, raising the question of whether they share an underlying...

https://arxiv.org/abs/2610.32281

7 CONF

PASTA: Prompt-Aware Annealed Diffusion Sampling

ArXiv CS.AI 50%

arXiv:2610.32390v1 Announce Type: cross Abstract: While Diffusion models produce high-quality images, real-world visual design requires intentionally preserving clean regions for headlines, logos, or UI elements. Existing layout-aware text-to-image methods struggle to strictly...

https://arxiv.org/abs/2610.32390

8 CONF

DIGOM: Disentangled Interaction Graphs for Outcome prediction in Multimodal healthcare

ArXiv CS.AI 50%

arXiv:2610.32451v1 Announce Type: cross Abstract: Multimodal healthcare data contain relational information that extends beyond individual patient observations. Effective clinical outcome prediction therefore requires not only integrating information from multiple modalities,...

https://arxiv.org/abs/2610.32451

9 CONF

Tool-Calling LLM Agents Under Realistic MCP Faults: Scale Does Not Purchase Fault Tolerance

ArXiv CS.AI 50%

arXiv:2610.32555v1 Announce Type: cross Abstract: LLM agents increasingly rely on external tools exposed via the Model Context Protocol (MCP), yet current benchmarks assume tools respond correctly and on time, leaving agent resilience to realistic tool failures untested. We...

https://arxiv.org/abs/2610.32555

0 CONF

Can AI Agents Help Robotic Engineers? Lessons from Two Focus Days in Industry

ArXiv CS.AI 50%

arXiv:2610.32636v1 Announce Type: cross Abstract: Software developers increasingly use AI agents that execute multi-step programming tasks on their own. However, it is unclear if and how this transfers to robotic engineering, where engineers program industrial robots in niche,...

https://arxiv.org/abs/2610.32636

1 CONF

InferSlice: A Reference Architecture for SLO-driven LLM Inference over a Telco Wide-Area Network

ArXiv CS.AI 50%

arXiv:2610.32638v1 Announce Type: cross Abstract: LLM inference is increasingly deployed as a core service for enterprise and interactive applications, but sovereignty, compliance, latency and resource-locality requirements can limit the suitability of centralised cloud...

https://arxiv.org/abs/2610.32638

2 CONF

AI Alignment for Digital Democracy: Fairer Voters Representation against LLM Biases

ArXiv CS.AI 50%

arXiv:2610.32780v1 Announce Type: cross Abstract: AI representation is emerging in digital democracy, assisting in voter emulation and decision-making. Large language models (LLMs) behave differently depending on the model, the prompt context, and the voter information they...

https://arxiv.org/abs/2610.32780

3 CONF

PoKER: A Probability of Kill Estimation Rate Model for Air-to-Air Missiles Using Machine Learning on Stochastic Targets

ArXiv CS.AI 50%

arXiv:2610.32864v1 Announce Type: cross Abstract: This work introduces PoKER, a novel probabilistic model engineered to optimize missile launch effectiveness in air-to-air scenarios, specifically within Beyond Visual Range (BVR) air combat. Unlike conventional Weapon Engagement...

https://arxiv.org/abs/2610.32864

4 CONF

Interpretable sparse fine-tuning with TopKLoRA enables faithful circuit discovery

ArXiv CS.AI 50%

arXiv:2610.32866v1 Announce Type: cross Abstract: Fine-tuning on datasets of unknown origin can create a backdoor in the trained model when the data includes poisoned examples. Difficult to localise and remove, backdoors can persist through safety training. While...

https://arxiv.org/abs/2610.32866

5 CONF

Green Code or Costly Code? Evaluating the Energy Costs of Iterative LLM-Based Code Optimizations

ArXiv CS.AI 50%

arXiv:2610.32891v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly used to optimize code for performance. This practice often assumes that improvements in execution speed naturally translate into overall lower energy consumption. We challenge this...

https://arxiv.org/abs/2610.32891

6 CONF

Yalda-Embedding: Open Development of a Competitive Persian Retrieval Model

ArXiv CS.AI 50%

arXiv:2610.32933v1 Announce Type: cross Abstract: Despite advances in multilingual text representations, effective dense retrieval for Persian remains constrained by the limited availability of language-specific training resources and compact embedding models. We present Yalda...

https://arxiv.org/abs/2610.32933

7 CONF

Monetizing Intelligence: Advertising and Subscription Design on LLM Platforms

ArXiv CS.AI 50%

arXiv:2610.32945v1 Announce Type: cross Abstract: Large language model (LLM) platforms introducing advertising must decide how much capability ad revenue should fund and what consumers should pay to avoid ads. We study a monopolistic platform that jointly chooses costly response...

https://arxiv.org/abs/2610.32945

8 CONF

MEDMKG: Benchmarking Medical Knowledge Exploitation with Multimodal Knowledge Graph

ArXiv CS.AI 50%

arXiv:2505.17214v2 Announce Type: replace Abstract: Medical deep learning models depend heavily on domain-specific knowledge to perform well on knowledge-intensive clinical tasks. Prior work has primarily leveraged unimodal knowledge graphs, such as the Unified Medical Language...

https://arxiv.org/abs/2505.17214

9 CONF

Testing Transformer Learnability on the Arithmetic Sequence of Rooted Trees

ArXiv CS.AI 50%

arXiv:2512.01870v3 Announce Type: replace Abstract: We study whether a transformer network can learn the deterministic sequence of trees generated by the iterated prime factorization of the natural numbers. Each integer is mapped into a rooted planar tree and the resulting...

https://arxiv.org/abs/2512.01870

0 CONF

MedInsightBench: Evaluating Medical Analytics Agents Through Multi-Step Insight Discovery in Multimodal Medical Data

ArXiv CS.AI 50%

arXiv:2512.13297v2 Announce Type: replace Abstract: In medical data analysis, extracting deep insights from complex, multi-modal datasets is essential for improving patient care, increasing diagnostic accuracy, and optimizing healthcare operations. However, there is currently a...

https://arxiv.org/abs/2512.13297

1 CONF

Spatial disparities in simulated patrol detection: a reproducible generative-model audit across Baltimore and Chicago

ArXiv CS.AI 50%

arXiv:2603.18987v3 Announce Type: replace Abstract: Auditing predictive policing requires distinguishing recorded crime incidents, simulated patrol observations, and the demographic characteristics of places. We present a reproducible spatial simulation audit combining a...

https://arxiv.org/abs/2603.18987

2 CONF

Architecture-Aware Reinforcement Learning Makes Sliding-Window Attention Competitive in Math Reasoning

ArXiv CS.AI 50%

arXiv:2606.11634v2 Announce Type: replace Abstract: The rapid progress of reasoning and agentic large language models (LLMs) has increased the demand for long-context inference, but self-attention (SA) scales quadratically with context length. To address this, we study SWARR...

https://arxiv.org/abs/2606.11634

3 CONF

Attractor Domain Theory: A Mathematical Framework for Cardiovascular Attractor Analysis with Wearable Photoplethysmography (PPG) Validation

ArXiv CS.AI 50%

arXiv:2606.22039v3 Announce Type: replace Abstract: The cardiac attractor provides a phase-space representation of cardiovascular dynamics reconstructed from scalar observations under standard delay-embedding conditions. We introduce Attractor Domain Theory (ADT), a theory in...

https://arxiv.org/abs/2606.22039

4 CONF

Where Cross-Question Structure in LLM Errors Comes From: Shared Seeds, Shared Prompts, and Answer-Letter Preference

ArXiv CS.AI 50%

arXiv:2607.20464v2 Announce Type: replace Abstract: Disagreement between language models is often read as missing knowledge. Questions that need the same missing knowledge should then be right or wrong together. We test for this structure with the eigenvalues of the questions'...

https://arxiv.org/abs/2607.20464

5 CONF

AI-guided deep visual proteomics reveals recurrence-associated morpho-molecular signatures within triple-negative breast cancers

ArXiv CS.AI 50%

arXiv:2608.03145v2 Announce Type: replace Abstract: Outcome informed AI can localize prognostically relevant regions in digital pathology, but directly characterizing the molecular basis of these spatial predictions remains challenging. Here, we developed an outcome informed,...

https://arxiv.org/abs/2608.03145

6 CONF

Interchangeable in Role, Not in History: Replacing Members of Formed LLM Teams

ArXiv CS.AI 50%

arXiv:2609.05279v2 Announce Type: replace Abstract: Multiagent LLM systems often treat a role as a slot: when a member is replaced, any agent competent at the role is expected to fill it. We test this view on teams whose members keep short written notebooks. A one-seat swap...

https://arxiv.org/abs/2609.05279

7 CONF

When Clinical Judgment Becomes a Predictive Shortcut: Two Questions Clinical AI Must Answer Before Deployment

ArXiv CS.AI 50%

arXiv:2609.25852v2 Announce Type: replace Abstract: Clinical AI can predict outcomes using information generated by existing clinical judgment. Such performance alone cannot establish usefulness before recognition or across care processes. We propose evidence requirements...

https://arxiv.org/abs/2609.25852

8 CONF

Symbolic Guidance for LLM Agents in Distributed Multiagent Coordination

ArXiv CS.AI 50%

arXiv:2609.31963v2 Announce Type: replace Abstract: Large language models (LLMs) are increasingly deployed as autonomous agents in multi-agent systems, yet their ability to reliably execute distributed coordination protocols remains poorly understood. While AgentsNet, a...

https://arxiv.org/abs/2609.31963

9 CONF

Train4Merge: RL or SFT---Which Yields Better Teachers for OPD-Based Model Merging?

ArXiv CS.AI 50%

arXiv:2609.32303v3 Announce Type: replace Abstract: Domain experts trained from a shared checkpoint can transfer their specialized capabilities to a single student through on-policy distillation (OPD). Existing research primarily focuses on improving this merging process, while...

https://arxiv.org/abs/2609.32303

00 CONF

Accelerating Wildfire Hazard Mapping Across Canada with an AI Surrogate Modelling Framework

ArXiv CS.AI 50%

arXiv:2610.30170v2 Announce Type: replace Abstract: Canada's increasing wildfire activity requires frequent national hazard mapping to guide proactive mitigation, including fuel management, land-use planning, infrastructure protection, and evacuation routing. However,...

https://arxiv.org/abs/2610.30170

01 CONF

Mitigating Saliency Collapse: Robust Saliency-Aware Long-Text Image-Text Alignment

ArXiv CS.AI 50%

arXiv:2610.31005v2 Announce Type: replace Abstract: Long-text image-text alignment is essential for fine-grained scene understanding, yet existing methods mainly focus on extending context length or improving local alignment. We identify an overlooked failure mode, saliency...

https://arxiv.org/abs/2610.31005

02 CONF

On Orthogonal Recurrent Neural Networks Quantization

ArXiv CS.AI 50%

arXiv:2402.04012v3 Announce Type: replace-cross Abstract: In recent years, Orthogonal Recurrent Neural Networks (ORNNs) have gained popularity due to their ability to manage tasks involving long-term dependencies, such as the Copy-task, and their linear complexity. However,...

https://arxiv.org/abs/2402.04012

03 CONF

Power- and Fragmentation-aware Online Scheduling for GPU Datacenters

ArXiv CS.AI 50%

arXiv:2412.17484v2 Announce Type: replace-cross Abstract: The rise of Artificial Intelligence and Large Language Models is driving increased GPU usage in data centers for complex training and inference tasks, impacting operational costs, energy demands, and the environmental...

https://arxiv.org/abs/2412.17484

04 CONF

Enhancing Table Recognition with Vision LLMs: A Benchmark and Neighbor-Guided Toolchain Reasoner

ArXiv CS.AI 50%

arXiv:2412.20662v4 Announce Type: replace-cross Abstract: Pre-trained foundation models have recently made significant progress in table-related tasks such as table understanding and reasoning. However, recognizing the structure and content of unstructured tables using Vision...

https://arxiv.org/abs/2412.20662

05 CONF

Human-AI Interactions and the Psychological Triad: A Multi-Domain Survey of Cognitive, Behavioral, and Emotional Impacts

ArXiv CS.AI 50%

arXiv:2510.17753v3 Announce Type: replace-cross Abstract: As human-AI interactions increasingly reshape social and research landscapes, the risks of overreliance, cognitive offloading, and the nuanced degradation of human agency have become more apparent. This paper addresses...

https://arxiv.org/abs/2510.17753

06 CONF

Demo: Generative AI helps Radiotherapy Planning with User Preference

ArXiv CS.AI 50%

arXiv:2512.08996v3 Announce Type: replace-cross Abstract: Radiotherapy planning is a highly complex process that often varies significantly across institutions and individual planners. Most existing deep learning approaches for 3D dose prediction rely on reference plans as...

https://arxiv.org/abs/2512.08996

07 CONF

The Role of Quantum in Hybrid Quantum-Classical Neural Networks: An Empirical Assessment

ArXiv CS.AI 50%

arXiv:2601.04732v2 Announce Type: replace-cross Abstract: Quantum machine learning has emerged as a promising application domain for near-term quantum hardware, particularly through hybrid quantum-classical models that leverage both classical and quantum processing. Although...

https://arxiv.org/abs/2601.04732

08 CONF

PAIR-SAFE: A Paired-Agent Approach for Runtime Auditing and Refining AI-Mediated Mental Health Support

ArXiv CS.AI 50%

arXiv:2601.12754v2 Announce Type: replace-cross Abstract: LLMs are increasingly used for mental health support, yet they can produce responses that are overly directive, inconsistent, or clinically misaligned. Most risk mitigation approaches rely on implicit alignment through...

https://arxiv.org/abs/2601.12754

09 CONF

Contextual Memory Virtualisation: DAG-Based State Management and Structurally Lossless Trimming for LLM Agents

ArXiv CS.AI 50%

arXiv:2602.22402v2 Announce Type: replace-cross Abstract: As large language models engage in extended reasoning tasks, they accumulate significant state -- architectural mappings, trade-off decisions, codebase conventions -- within the context window. This understanding is lost...

https://arxiv.org/abs/2602.22402

10 CONF

MoE-SpAc: Efficient MoE Inference Based on Speculative Activation Utility in Heterogeneous Edge Scenarios

ArXiv CS.AI 50%

arXiv:2603.09983v2 Announce Type: replace-cross Abstract: Mixture-of-Experts (MoE) models enable scalable performance but face severe memory constraints on edge devices. Existing offloading strategies struggle with I/O bottlenecks due to the dynamic, low-information nature of...

https://arxiv.org/abs/2603.09983

11 CONF

Better Reward Models Are Not Always Less Biased: Social Alignment Beyond Standard Reward Benchmarks

ArXiv CS.AI 50%

arXiv:2605.05003v2 Announce Type: replace-cross Abstract: Reward models are central to LLM alignment, yet current benchmarks reveal little about how they represent diverse and sometimes competing social preferences. We extend reward model evaluation to four socially...

https://arxiv.org/abs/2605.05003

12 CONF

MedAction: Towards Active Multi-turn Clinical Diagnostic LLMs

ArXiv CS.AI 50%

arXiv:2605.07305v2 Announce Type: replace-cross Abstract: Most existing LLM diagnoses are evaluated on static, single-turn settings where complete patient information is provided upfront, an oversimplification of real clinical practice. We study active diagnosis: the real-life...

https://arxiv.org/abs/2605.07305

13 CONF

PolyMemDB: A Polyglot Database System for AI Memory Management

ArXiv CS.AI 50%

arXiv:2608.25577v2 Announce Type: replace-cross Abstract: With the widespread adoption of personal intelligent agents, users generate massive, heterogeneous data during long-term interactions. Leveraging this data as long-term memory helps reduce token overhead and deliver...

https://arxiv.org/abs/2608.25577

14 CONF

Training-Free Task Vectors for LLM Behavioral Control

ArXiv CS.AI 50%

arXiv:2609.09054v3 Announce Type: replace-cross Abstract: Task vectors enable post-training model editing by identifying semantically meaningful directions in weight space, typically computed as the difference between a fine-tuned model and its pretrained initialization....

https://arxiv.org/abs/2609.09054

15 CONF

Cross-Model In-Context Neurofeedback Reveals (Limited) Activation Control in LLMs

ArXiv CS.AI 50%

arXiv:2610.20179v2 Announce Type: replace-cross Abstract: Probes on activation vectors are widely used to detect misbehavior in language models, but their success relies on the assumption that models cannot control what their activations encode. Humans can learn, through...

https://arxiv.org/abs/2610.20179

16 CONF

GeoDiv KAN Solver for Time Fractional Navier Stokes Equations on Complex Domains to Realistic CFD Reconstruction

ArXiv CS.AI 50%

arXiv:2610.30179v2 Announce Type: replace-cross Abstract: Scientific machine learning provides a promising framework for solving complex problems in science and engineering. Notably, physics informed neural networks (PINNs) have shown considerable potential for solving...

https://arxiv.org/abs/2610.30179

17 CONF

OpenAttention: One Percent Is Enough with Tail Proxies

ArXiv CS.AI 50%

arXiv:2610.30951v2 Announce Type: replace-cross Abstract: Sparse attention reads only a small subset of the key-value cache, and how to treat the unread tail remains a key challenge. Discarding the tail biases the output, and sampling it as previous approaches do fails at reads...

https://arxiv.org/abs/2610.30951